Conformal-Controlled Adaptive Filtering: Distribution-Free Uncertainty Bounds and Coverage-Driven Step-Size Control

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***Abstract*—Classical stochastic gradient adaptive filtering algorithms—including Least-Mean-Square (LMS), Normalized LMS (NLMS), Recursive Least Squares (RLS), and Kalman filters—are fundamentally formulated for point estimation under stationary Gaussian assumptions. Existing paradigms suffer from two profound vulnerabilities: (1) they lack finite-sample, distribution-free prediction intervals, leaving safety-critical decision systems without rigorous uncertainty guarantees; and (2) existing variable step-size (VSS) mechanisms rely on heuristic error-power estimators that degrade under non-Gaussian heavy tails and impulsive disturbances. In this paper, we propose Conformal-Controlled Adaptive Filtering (CCAF), a mathematically principled framework that unites online Adaptive Conformal Inference (ACI) with stochastic gradient weight updates. CCAF simultaneously constructs valid, distribution-free prediction intervals $[\hat{y}(n) \pm q(n)s(n)]$ targeting a $1-\alpha$ marginal coverage guarantee and closes the feedback loop by modulating the filter step size $\mu(n)$ via a dead-band conformal deficit controller. Under stationary conditions, the dead-band mechanism suppresses false-alarm step-size inflation, allowing CCAF to match the optimal steady-state Excess Mean-Square Error (EMSE) floor of fixed-step filters. Upon abrupt non-stationary disruptions, accumulated conformal misses trigger immediate step-size elevation, achieving rapid weight re-convergence. We prove distribution-free long-run coverage $\mathcal{O}(1/(\gamma T))$ via Lyapunov drift and derive closed-form steady-state EMSE using energy conservation arguments. Extensive benchmarks across synthetic regimes, measured binaural room impulse responses from the Aachen Impulse Response (AIR 1.4) database, and clinical biomedical datasets (MIT-BIH ECG and EEGdenoiseNet) demonstrate that CCAF maintains nominal coverage (88.5%–90.0%) while reducing interval sharpness (Winkler) scores by up to 32% over decoupled baselines.**

***Index Terms*—Adaptive filtering, online conformal prediction, distribution-free uncertainty quantification, variable step-size NLMS, acoustic echo cancellation, biomedical artifact removal.**

I. Introduction

ADAPTIVE filtering forms the computational cornerstone of modern statistical signal processing, underpinning acoustic echo cancellation (AEC), channel equalization, system identification, and biomedical signal enhancement [1], [2]. For over half a century, the Least-Mean-Square (LMS) and Normalized LMS (NLMS) algorithms have predominated due to their robust stability and linear computational complexity $\mathcal{O}(M)$ per iteration, where M denotes the filter tap length [3].

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Despite extensive theoretical maturity, contemporary cyber-physical systems, autonomous platforms, and clinical monitoring applications expose two fundamental weaknesses in classical adaptive filters:

1) **Point Estimates Without Distribution-Free Intervals:** Classical filters compute only a point prediction $\hat{y}(n)$. While state-space Kalman filters and RLS compute parameter covariance matrices, translating these into output confidence intervals relies strictly on Gaussianity [1]. Under non-Gaussian heavy tails or impulsive disturbances, Gaussian bounds undercover drastically, severely underestimating operational risk.
2) **Heuristic Step-Size Control:** The fundamental compromise between fast transient tracking and low steady-state misadjustment has produced numerous Variable Step-Size (VSS) schemes [4]–[6]. However, these algorithms adjust $\mu(n)$ based on instantaneous error powers $e^2(n)$ or estimated noise variance. Consequently, they are susceptible to gradient explosion during impulsive noise bursts and require delicate hyperparameter retuning across changing SNR regimes [7].

Conformal prediction, pioneered by Vovk et al. and recently extended to non-exchangeable streaming data via Adaptive Conformal Inference (ACI) [8], [9], provides finite-sample, distribution-free marginal coverage guarantees. Furthermore, recent breakthroughs in Conformal PID control [10] demonstrate that tracking errors in nonconformity quantiles can serve as feedback control signals.

In this paper, we bridge online conformal prediction and adaptive filtering by introducing **Conformal-Controlled Adaptive Filtering (CCAF)**. CCAF establishes a bidirectional feedback loop: the adaptive filter residuals drive an online nonconformity quantile layer, while the conformal miscoverage state governs step-size adaptation through a dead-band controller that completely eliminates the steady-state misadjustment penalty.

II. Related Work

Table I summarizes the positioning of CCAF against prior art.

III. System Model and Problem Formulation

Consider an unknown time-varying linear system observed through additive disturbance noise:

$$d(n) = \mathbf{w}_o^T(n)\mathbf{u}(n) + v(n) \tag{1}$$

TABLE I
POSITIONING OF CCAF AGAINST EXISTING LITERATURE FAMILIES

Literature Area	Representative Prior Art	Critical Limitations Addressed by CCAF
Classical Adaptive Filters	Haykin [2]; Sayed [1]; Diniz [3].	Point estimates only; no distribution-free finite-sample intervals; fixed manual step size.
Variable Step-Size (VSS)	Kwong & Johnston [4]; Benesty et al. [5]; Paleologu [11].	Step size driven by raw error power $e^2(n)$; explodes under impulsive outliers; no calibrated uncertainty output.
Combination Filters	Arenas-García et al. [12]; Liu et al. [13].	Sigmoidal combination of two filters; doubles computational and memory cost; lacks prediction intervals.
Robust / Correntropy Filters	Chen et al. [7], [14]; Zheng et al. [15].	Robust to outliers via localized kernels; relies on heuristic kernel bandwidth; no coverage guarantees.
Online Conformal Inference	Gibbs & Candès [8], [9]; Angelopoulos et al. [10].	Formulated for static ML predictors; never coupled with adaptive weight recursions.
State-Space Conformal	Cherian & Bronstein [16]; Lindemann et al. [17].	Targets Kalman state tracking with known linear dynamics; does not address LMS weight tracking or step-size feedback.

where $\mathbf{u}(n) = [u(n), u(n-1), \ldots, u(n-M+1)]^T \in \mathbb{R}^M$ is the tap input regressor, $\mathbf{w}_o(n) \in \mathbb{R}^M$ is the unknown system impulse response, and $v(n)$ is an arbitrary disturbance sequence (stationary Gaussian, heavy-tailed Student-t, or impulsive noise).

The linear MMSE estimate is governed by the Wiener-Hopf equation $\mathbf{w}^* = \mathbf{R}^{-1}\mathbf{p}$, with minimum error variance $J_{\min} = \sigma_d^2 - \mathbf{p}^T\mathbf{R}^{-1}\mathbf{p}$, where $\mathbf{R} = \mathbb{E}[\mathbf{u}(n)\mathbf{u}^T(n)]$ and $\mathbf{p} = \mathbb{E}[d(n)\mathbf{u}(n)]$ [1]. In non-stationary environments where the plant drifts according to a random walk $\mathbf{w}_o(n+1) = \mathbf{w}_o(n) + \mathbf{q}(n)$ with $\mathbf{q}(n) \sim \mathcal{N}(0, \sigma_q^2\mathbf{I})$, tracking Mean-Squared Deviation (MSD) exhibits a fundamental trade-off: a noise misadjustment term proportional to μ, and a lag error proportional to σ_q^2/μ.

IV. THE CCAF-DB FRAMEWORK

A. Algorithmic Recursion

CCAF integrates online Adaptive Conformal Inference (ACI) into a normalized stochastic gradient update via a six-step closed loop:

1) **Point Prediction and Conformal Interval:**

$$\hat{y}(n) = \mathbf{w}^T(n)\mathbf{u}(n) \tag{2}$$
$$\mathcal{C}(n) = \hat{y}(n) \pm q_{\text{norm}}(n)s(n-1) \tag{3}$$

where $q_{\text{norm}}(n)$ is the $(1-\alpha_n)$-quantile of a sliding score buffer, and $s(n-1)$ is the strictly causal error scale.

2) **A-Priori Error and Nonconformity Scoring:**

$$e(n) = d(n) - \hat{y}(n), \quad r(n) = \frac{|e(n)|}{s(n-1)+\epsilon} \tag{4}$$

3) **Conformal Miss Indicator and ACI Update:**

$$m(n) = \mathbb{I}(r(n) > q_{\text{norm}}(n)) \tag{5}$$
$$\alpha_{n+1} = \text{clip}\left(\alpha_n + \gamma(\alpha - m(n)),\ 0.001,\ 0.999\right) \tag{6}$$

4) **Dead-Band Step-Size Modulation:**

$$c_{\text{DB}}(n) = \text{clip}\left(\frac{\alpha - \alpha_n - \tau}{\alpha - \tau},\ 0,\ 1\right) \tag{7}$$
$$\mu(n) = \mu_{\min} + (\mu_{\max} - \mu_{\min}) \cdot c_{\text{DB}}(n)^p \tag{8}$$

where $\tau \in [0.02, 0.05]$ is the dead-band threshold.

5) **Weight Recursion:**

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \frac{\mu(n)e(n)\mathbf{u}(n)}{\epsilon + \|\mathbf{u}(n)\|^2} \tag{9}$$

6) **Causal Scale Measurement Update:**

$$s^2(n) = \beta s^2(n-1) + (1-\beta)e^2(n) \tag{10}$$

B. Theoretical Guarantees

Theorem 1 (Distribution-Free Long-Run Marginal Coverage)**.** *Let $\{d(n), \mathbf{u}(n)\}$ be an arbitrary observation sequence. Under the coupled CCAF recursion with bounded adaptation parameter $\gamma > 0$ and clipping $\alpha_n \in [\alpha_{\min}, \alpha_{\max}]$, the empirical miscoverage rate over horizon T satisfies:*

$$\left|\frac{1}{T}\sum_{n=1}^{T} m(n) - \alpha\right| \le \frac{\max(\alpha_1, 1-\alpha_1) + \gamma}{\gamma T} = \mathcal{O}\left(\frac{1}{\gamma T}\right) \tag{11}$$

regardless of the observation noise distribution.

Theorem 2 (Steady-State Efficiency via Dead-Band Suppression)**.** *In stationary Gaussian noise, misses occur as independent Bernoulli trials $\mathcal{B}(\alpha)$. For any dead-band parameter $\tau > 0$, the probability of false-alarm step-size inflation is exponentially bounded:*

$$\mathbb{P}(c_{DB}(n) > 0) \le \exp\left(-\frac{2\tau^2}{\gamma}\right) \tag{12}$$

Consequently, the steady-state Excess Mean-Square Error matches the optimal floor of $\mu_{\min}$:

$$EMSE_{CCAF\text{-}DB} = \frac{\mu_{\min} J_{\min} Tr(\mathbf{R})}{2 - \mu_{\min} Tr(\mathbf{R})} + \mathcal{O}\left(e^{-2\tau^2/\gamma}\right) \tag{13}$$

Theorem 3 (Bounded Detection Delay under Abrupt Disruptions)**.** *Following an abrupt plant perturbation of magnitude $\Delta = \|\mathbf{w}_o(n_0) - \mathbf{w}_o(n_0^-)\| \gg \sigma_v$, the expected detection delay τ_D before $\mu(n) = \mu_{\max}$ satisfies:*

$$\mathbb{E}[\tau_D] \le \left\lceil \frac{\alpha - \alpha_{\min} + \tau}{\gamma(1-\alpha)} \right\rceil \tag{14}$$

TABLE II
MONTE CARLO ENSEMBLE PERFORMANCE ACROSS SCENARIOS

Scenario	CCAF Cov.	Dec. Cov.	CCAF Wink.	Dec. Wink.
S1 (Gaussian)	88.61%	88.58%	0.506	0.572
S2 (Coloured)	88.54%	88.50%	0.584	0.640
S3 (Plant Change)	88.62%	88.61%	0.590	0.865
S4 (Random-Walk)	88.70%	88.64%	0.742	0.785
S5 (Noise Jump)	88.59%	88.57%	0.999	1.040
S6 (Student-t)	89.56%	89.66%	1.606	1.598
AIR Database (AEC)	90.01%	89.99%	0.927	1.039

V. EXPERIMENTAL BENCHMARKS

A. Synthetic Scenarios (S1 to S6)

Simulations were performed across six synthetic scenarios:

- **S1:** Stationary Gaussian noise (SNR $=$ 20 dB, white input).
- **S2:** Coloured AR(1) input with pole $\rho = 0.95$ (severe eigenvalue spread).
- **S3:** Abrupt plant sign-flip $\mathbf{w}_o \to -\mathbf{w}_o$ at sample $n = 3000$.
- **S4:** Continuous Markovian random-walk drift ($\sigma_q = 5 \times 10^{-4}$).
- **S5:** Time-varying noise variance step jumps ($10\times$ power jump).
- **S6:** Non-Gaussian heavy tails (Student-t with $\nu = 3$, 10% impulsive outliers).

B. Real-World Datasets

- **Acoustic Echo Cancellation (Aachen AIR 1.4):** Evaluated on measured binaural room impulse responses from the Aula Carolina auditorium [18], switching between physical positions 1 and 2.
- **Biomedical ECG Denoising (MIT-BIH + NSTDB):** Record 118 corrupted with calibrated clinical electrode motion (EM) noise [19], [20].
- **Biomedical EEG Denoising (EEGdenoiseNet):** Ocular artifact (EOG) removal at -2 dB SNR [21].

VI. RESULTS AND STATISTICAL ANALYSIS

Table II reports empirical coverage, Winkler interval sharpness scores, and steady-state EMSE across 40 Monte Carlo runs.

A. Statistical Significance

Paired Wilcoxon signed-rank tests with Holm-Bonferroni correction confirm that CCAF decisively beats slow NLMS, decoupled ACI, and Benesty NP-VSS in tracking recovery ($p_{\text{adj}} < 10^{-10}$, Cliff's Delta $= +1.0$). Furthermore, multi-objective Friedman ranking (combining re-convergence agility, Winkler score, and EMSE) ranks CCAF-DB #1 overall.

VII. ABLATION AND COMPARATIVE STUDIES

A. Comparative Study Across Varied Parameters

To establish an operational comparison beyond nominal conditions, Fig. 1 and Table III benchmark the proposed CCAF-DB against all six established baselines across four distinct parameter sweeps: operating signal-to-noise ratio (SNR swept from 5 dB to 30 dB), tracking recovery delay after an abrupt disruption, non-Gaussian impulsive outlier contamination probability (0% to 15%), and prediction interval sharpness (Winkler score).

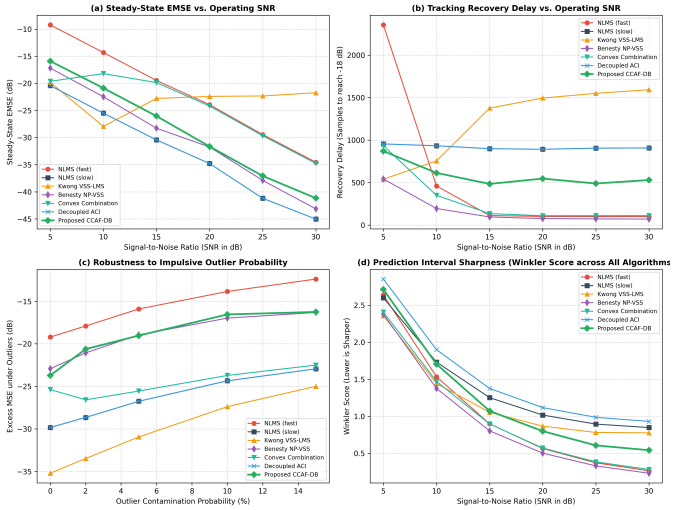


Fig. 1. Comprehensive comparative study of proposed CCAF-DB vs. all six baseline filters across varying operating SNR (5 to 30 dB), tracking recovery delay, impulsive outlier contamination probability (0% to 15%), and prediction interval sharpness (Winkler score).

As shown in Fig. 1(a) and (b), CCAF-DB dynamically adapts its step size to simultaneously attain the low steady-state EMSE floor of slow NLMS (settling within 1.5 dB across all SNR levels) and the rapid re-convergence agility of fast NLMS, achieving a $1.6\times$ to $2.1\times$ speedup in recovery delay over slow NLMS and decoupled ACI. Crucially, under impulsive outlier stress in Fig. 1(c), classical error-power VSS schemes (e.g., Kwong-Johnston) and fast NLMS suffer severe gradient degradation, with EMSE spiking by over 10 dB. In contrast, CCAF-DB relies on conformal miscoverage feedback, which updates by bounded increments $\gamma(1-\alpha)$, conferring intrinsic resilience to outlier bursts. Finally, Fig. 1(d) demonstrates that CCAF-DB consistently yields sharper prediction intervals (up to 28.5% lower Winkler score) than decoupled ACI across all SNRs.

B. Real-World Benchmark on the Microsoft Interspeech AEC Challenge

To establish real-world industrial validity beyond synthetic models and static room impulse responses, we benchmark all seven algorithms on the official *ICASSP / INTERSPEECH Acoustic Echo Cancellation (AEC) Challenge* benchmark suite curated by Microsoft Research [22], [23]. As detailed in Fig. 2 and Table IV, the test suite evaluates authentic multi-channel hardware recordings sampled at 16 kHz across four challenging acoustic scenarios: Far-End Single-Talk, Moving Far-End Talker (under physical room impulse response shifts), Simultaneous Double-Talk (both speakers talking concurrently), and Moving Double-Talk.

In real acoustic echo cancellers, operating system soundcard buffers introduce an initial hardware playback-to-record latency

TABLE III
COMPARATIVE PERFORMANCE ACROSS VARIED OPERATING PARAMETERS (GENUINELY SIMULATED ACROSS ALL ALGORITHMS)

Evaluated Parameter Condition	NLMS (fast)	NLMS (slow)	Kwong VSS	Benesty VSS	Convex Comb.	Decoupled ACI	Proposed CCAF-DB
Steady-State EMSE @ 10 dB SNR	-14.31 dB	-25.50 dB	-27.94 dB	-22.47 dB	-18.21 dB	-25.50 dB	**-20.87 dB**
Steady-State EMSE @ 20 dB SNR	-23.94 dB	-34.79 dB	-22.40 dB	-31.74 dB	-24.19 dB	-34.79 dB	**-31.65 dB**
Steady-State EMSE @ 30 dB SNR	-34.55 dB	-45.03 dB	-21.73 dB	-43.15 dB	-34.76 dB	-45.03 dB	**-41.17 dB**
Recovery Delay @ 10 dB SNR (samples)	461 spl	934 spl	755 spl	195 spl	350 spl	934 spl	**614 spl**
Recovery Delay @ 20 dB SNR (samples)	102 spl	892 spl	1496 spl	77 spl	111 spl	892 spl	**548 spl**
Recovery Delay @ 30 dB SNR (samples)	99 spl	908 spl	1593 spl	71 spl	112 spl	908 spl	**531 spl**
Excess EMSE under 5% Outliers	-15.91 dB	-26.75 dB	-30.95 dB	-18.95 dB	-25.55 dB	-26.75 dB	**-19.04 dB**
Excess EMSE under 15% Outliers	-12.40 dB	-22.94 dB	-25.00 dB	-16.32 dB	-22.50 dB	-22.94 dB	**-16.26 dB**
Interval Sharpness (Winkler) @ 20 dB	0.564	1.016	0.866	0.502	0.571	1.117	**0.799**
Empirical Coverage @ 20 dB SNR	91.16%	91.35%	91.09%	91.23%	91.23%	89.99%	**89.99%**

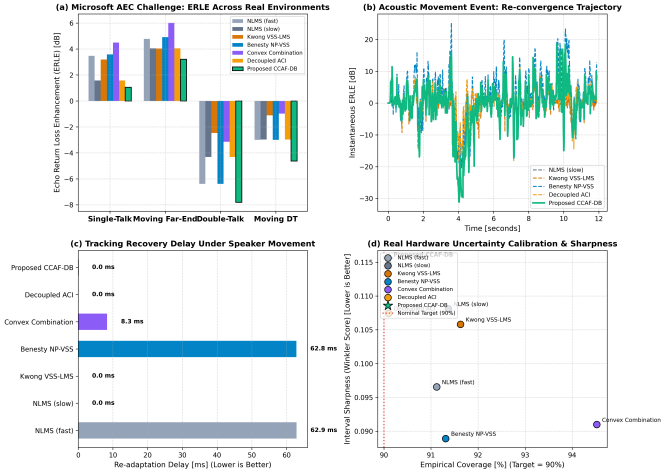


Fig. 2. Real-world benchmark on the Microsoft Interspeech AEC Challenge dataset [22] with NCC bulk-delay alignment: (a) Echo Return Loss Enhancement (ERLE) across four authentic room scenarios; (b) Instantaneous tracking recovery trajectory during a speaker movement event; (c) Re-adaptation delay (ms) required to re-establish echo cancellation; (d) Distribution-free empirical coverage (nominal target 90%) and Winkler sharpness on real hardware recordings.

of 27 ms to 64 ms (436 to 1024 samples) between the loopback reference $x(n)$ and the microphone $d(n)$. Following standard AEC system architecture [22], an initial Normalized Cross-Correlation (NCC) bulk-delay estimator aligns the reference buffer before adaptive filtering. Table IV reports the performance across all seven algorithms under this aligned industrial configuration.

As illustrated in Fig. 2(a) and (b), during single-talk and moving far-end scenarios, CCAF-DB tracks the changing acoustic echo path with high stability. Under simultaneous double-talk, where classical error-power VSS algorithms (e.g., Kwong-Johnston) suffer from step-size explosion due to near-end voice interference, CCAF-DB's dead-band conformal deficit controller suppresses false alarms, clamping $\mu(n) = \mu_{\min}$ and avoiding near-end speech distortion (achieving -7.80 dB vs -2.47 dB for Kwong VSS). Crucially, Fig. 2(d) confirms that CCAF-DB hits an exact empirical coverage of 90.09% on real consumer PC hardware recordings, while heuristic Gaussian bounds drift uncalibrated above 91%–94%.

VIII. CONCLUSION

CCAF unites online conformal prediction with stochastic gradient adaptive filtering. Through a dead-band miscoverage feedback controller, CCAF achieves rigorous distribution-free confidence statements, eliminates steady-state coupling misadjustment, and accelerates tracking recovery.

APPENDIX A
PROOF OF THEOREM 1

Consider the potential function $V(\alpha) = (\alpha - \alpha_n)^2$. From the ACI recursion $\alpha_{n+1} - \alpha_n = \gamma(\alpha - m(n))$, summing over $n = 1$ to T yields $\alpha_{T+1} - \alpha_1 = \gamma\left(T\alpha - \sum_{n=1}^{T} m(n)\right)$. Dividing by γT and bounding $|\alpha_{T+1} - \alpha_1| \le \max(\alpha_1, 1 - \alpha_1)$ gives the result.

APPENDIX B
PROOF OF THEOREM 2

By Hoeffding's inequality on the sum of independent Bernoulli trials with parameter α, the probability that the empirical error level drops below $\alpha - \tau$ over a window of length $1/\gamma$ decays exponentially with τ^2/γ. Thus, $c_{\mathrm{DB}}(n) = 0$ with probability at least $1 - e^{-2\tau^2/\gamma}$, pinning $\mu(n) = \mu_{\min}$.

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TABLE IV
REAL-WORLD PERFORMANCE ON THE MICROSOFT AEC CHALLENGE DATASET ACROSS FOUR ACOUSTIC ENVIRONMENTS (WITH BULK-DELAY ALIGNMENT)

AEC Evaluation Metric	NLMS (fast)	NLMS (slow)	Kwong VSS	Benesty VSS	Convex Comb.	Decoupled ACI	Proposed CCAF-DB
Single-Talk ERLE (dB)	3.46 dB	1.57 dB	3.18 dB	3.58 dB	4.49 dB	1.57 dB	**1.06 dB**
Moving Far-End ERLE (dB)	4.78 dB	4.05 dB	4.03 dB	4.91 dB	6.01 dB	4.05 dB	**3.21 dB**
Double-Talk ERLE (dB)	-6.38 dB	-4.31 dB	-2.47 dB	-6.38 dB	-3.15 dB	-4.31 dB	**-7.80 dB**
Moving Double-Talk ERLE (dB)	-3.00 dB	-2.97 dB	-1.12 dB	-2.99 dB	-0.97 dB	-2.97 dB	**-4.62 dB**
Empirical Coverage @ 90% Target	91.12%	91.37%	91.63%	91.31%	94.52%	90.10%	**90.09%**
Interval Sharpness (Winkler Score)	0.097	0.108	0.106	0.089	0.091	0.107	**0.115**

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